

AutoTrade — an autonomous AI trading bot, built to test itself

A self-running trading system for US equities & defined-risk options, engineered around one unusual idea: don't assume the AI has an edge — **measure** whether it does, and distill the part that works into code.

WHAT IT IS

A bot that trades on its own. Each qualifying cycle, an AI model (Claude) proposes **one** trade; a deterministic code layer can veto, resize, or override it — but can never invent a trade. It runs unattended on a schedule, manages its own positions, enforces its own risk limits, and reports to the operator by phone (Telegram). **Paper-traded only** — no live-money risk — while its edge is validated.

DESIGN PHILOSOPHY — THE MODEL PICKS, THE CODE PROTECTS

Most AI-trading projects hand the model the keys. This one inverts the trust: the LLM is wrapped in layers of deterministic guardrails it cannot override.

- **Regime gate** — market trend × volatility decide which strategies may even open (or force cash).
- **Event router** — rides fresh catalysts; braces before scheduled binary events.
- **Guardrails** — anti-chase, counter-trend blocks, per-name & correlation caps, a hard daily-loss halt.
- **Code-enforced exits** — trailing stops, breakeven locks, sector-confluence stops, **defined-risk only**.

The model only fills in a decision inside a box the code has already drawn.

WHAT MAKES IT DIFFERENT — IT MEASURES ITSELF

The hard question most projects skip: *does the AI's stock-picking actually add value, or is it expensive noise?* This bot is built to answer that with data:

- Every decision logs the **full choice set** — every candidate it saw, the features, the news it read, the pick, and the result.
- A research pipeline computes the **forward return of every candidate it passed on**, not just what it traded — the only honest way to measure selection skill.
- A **distillation** layer mines that data for a deterministic signal and tests whether the AI adds anything *beyond* the observable features.

The endgame: *distill the AI's useful behavior into deterministic code, and remove the AI wherever a rule reproduces it*. Model as *teacher*, not trader.

HONEST STATUS

- ✔ It runs autonomously, safely, and exactly as designed — **the engineering works**.
- ⚠ **Edge: unproven.** Three independent checks — a leak-free backtest, the academic literature, and the bot's *own* live decisions — so far show **no proven intraday selection edge**: the model's picks haven't beaten a random pick from the same candidate pool. The one durable signal found anywhere is long-only cross-sectional momentum — market *beta*, not alpha.
- 🔬 It is therefore run as an **instrumented experiment** — to find and distill an edge, not as a proven money-maker. Knowing *whether* it works is the entire point.

UNDER THE HOOD

Stack	Python · Claude (LLM) · Alpaca (broker + data) · scheduler · Telegram control
Self-learning	a nightly evidence-weighted rulebook — held <i>on trial</i> , measured not trusted
Risk	defined-risk structures only · hard daily-loss halt · paper account hardcoded · let winners run
Observability	localhost dashboard (live equity, decisions, vetoes, learnings) + full per-cycle logs

Research / validation phase. Paper-trading only. Built for rigor over hype — the goal is to know the truth about whether an LLM can trade, and to keep only the part that demonstrably works.